STRELKA PRESS ПРЕСС

THE ACTION IS THE FORM

VICTOR HUGO'S TED TALK

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Published by Strelka Press
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First edition. Published on Amazon and iBookstore.
ISBN 978-5-9903364-6-9

The Action is the Form: Victor Hugo's TED Talk

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Microwaves bounce between cell phones. Credit cards — all 0.76 mm thick — slip through the slots in cash machines anywhere in the world. Computers synchronise. Shipping containers calibrate the global transportation and production of goods. Nearly identical buildings and urban arrangements proliferate globally. All these ubiquitous and seemingly innocuous features of our world are evidence of global infrastructure.

The word "infrastructure" typically conjures associations with physical networks for transportation, communication or utilities — a hidden substrate or binding medium. Yet the technologies comprising these networks consist not only of underground grids of pipes and wires or tangles of fibre-optic cable on the bottom of the ocean, but also pools of microwaves beaming from satellites, atomised populations of electronic devices and shared technical platforms. Far from hidden, infrastructure is often the overt point of contact and access, where the underlying rules of the world can be clasped in the space of everyday life.

Making another, perhaps more important observation, buildings, and even whole cities, have become infrastructural technologies. From the fields of repeatable suburban houses and traffic-engineered highways of the mid 20th century to the malls, resorts, golf courses and big-box stores of contemporary culture, repeatable formulas make most of the space in the world. These buildings are not singularly crafted enclosures, but reproducible products — spatial products. The discipline of architecture is only responsible for a trickle of the world's spaces while a fire hose blasts out the rest. A familiar confetti of brightly coloured boxes nestled in black asphalt and bright green grass tells elaborate stories about Starbucks coffee, Beard Papa cream puffs and Arnold Palmer golf communities. This all too visible cartoon of abstract logics shapes most of the space in which we are swimming. Now not only small communities and resorts but also entire world cities are constructed according to a formula, usually a formula that replicates Shenzhen or Dubai anywhere in the world.

However familiar this mise-en-scène, popular culture has not yet found a compelling way to express the collapse between object and background. Some essential distinction between what is positive object and what is matrix must be dissolved: infrastructure is not just the urban substructure, but the urban structure itself — the very parameters of global urbanism. We do not build cities by accumulating singular masterpiece buildings. The constant flow of spatial products and urban formulas is more infrastructural. Architecture is making the occasional stone in the water. The world is making the water.

It is almost as if Victor Hugo would have the makings of a really good TED talk.

In *Notre Dame de Paris*, a 19th century novel set in the 15th century, Victor Hugo famously observed that "… architecture [like that of the cathedral] was developed in proportion with human thought; it became a giant with a thousand heads and a thousand arms, and fixed all this floating symbolism in an eternal, visible, palpable form." But speaking through the character of an archdeacon in the novel, he also predicted that Gutenberg's new technology threatened that giant. The printed word would usurp architecture as the vessel of cultural imagination and steal its supernatural power: "This will kill that. The book will kill the edifice."

At first, the contemporary evidence of urban space as an infrastructural technology might seem to confirm Hugo's assertion about the death of architecture. And there is no doubt of an ongoing textual information explosion. Yet, at this juncture, Hugo has a chance to step into the 21st century and make an astonishing reversal. He can recuperate the power of space as the carrier of an unspoken, undeclared cultural imaginary. He can demonstrate that the mystifying giant with a thousand heads and thousand arms is alive again, in the explosive growth of a heavy, material, non-textual medium: the matrix space of global infrastructure. The new giant is also the secret weapon of the most powerful people in the world. It cannot be petted or tamed, but it can be manoeuvred and exploited. Doing so requires a political art — an art found in what was presumed to be an artless background. Spatial technologies might even have the power and currency of, not text, but software: an updating platform for shaping the city. Hugo could author another cultural meme on the order of "this kills that". Architecture, killed by the book, is reincarnate as something more powerful still — *as information itself*.

It would take a bit of handling from the TED people. Hugo would have to be a little more upbeat. It's the *disposition* of the performance that matters — who says it, how they say it, who repeats it and to whom. This entrepreneurial Hugo wouldn't be talking to an audience of architects. The 19th-century Hugo provided architecture with a staple of rhetorical angst about its lost mystical powers, and the discipline has nearly worn out this section of his novel with hand-wringing and soul searching. No, the 21st-century Hugo, if he is properly coached, is talking to another audience, angling for the elite players in the room who, perhaps because they are not involved with architecture culture, really understand something about the power of space. Architectural arts are the very thing that is needed and yet part of the riddle of Hugo's performance is why he must step away from architecture in order to return to it with a more robust audience.

The most ordinary spaces have to be rendered magical, but Hugo can do magical. It's why he would keep the beard. "Architecture is information itself". It needs a lot more explanation, but it sounds good. Inscrutability can work. As long as he has a body mic and a publicist, Hugo can aim to be the guru who is a little bit cryptic — or even a little bit florid and evangelical. "Impassioned" will be the word to use in promotional blurbs and dust jacket copy. It sells, and a hush will fall over his various audiences as they creep up to his knee to receive imponderables. "Architecture is information". Hugo becomes an industry. The TED digital flourish.

The giant is global infrastructure space.

The TED audience will recognize Hugo's supernatural giant with a thousand heads and a thousand arms as an apt model for the role of space in global politics. Some of the most radical changes to the globalizing world are being made, not in the language of law and diplomacy, but rather by contagious spatial formulas. Often at a remove from familiar legislative processes, these infrastructures generate *defacto* forms of polity faster than official forms of governance can legislate them.

For example, the infrastructural model for Dubais and Shenzhens — the free trade zone — provides one glimpse of the giant. In the early 20th century, the free trade zone was a fenced compound for storing custom-free goods. As those compounds began to incorporate manufacturing, the United Nations Industrial Development Organisation began to promote the form as an industrial installation to kick-start the economies of developing countries. With administrations that are separate from their host state, the zone offers exemptions from taxes, labour laws or environmental regulations. While the exemptions were designed to avoid local bureaucracy, soon every corporation and every urban function wanted in. As a test of free market principles, China adopted the form for an entire city, first and most notably in Shenzhen, and incentivised urbanism has since become a global addiction. HITEC City in Hyderabad or King Abdullah Economic City in Saudi Arabia join scores of other similar zone "cities" around the world. Many adorn their corporate office parks with glittering skyscrapers and ecstatic signals of national pride as they celebrate entry into a network of similar zones. Growing exponentially, zone cities appear in almost every country — some a few hectares, some a few kilometres in size. The zone has swallowed the city.

The zone is then the shibboleth of the global marketplace. It is the perfect vessel for corporate "externalizing" — the means by which corporations eliminate obstacles to profit. Corporations like Halliburton, for instance, massage legislation in their home country but shelter from law by locating their headquarters in Dubai. While touted as a free market strategy, the zone is itself an instrument of market manipulation proffered by the "Washington Consensus" of the World Bank and the IMF. It is a suboptimal economic instrument, but the zone is so popular that major cities are developing their own zone doppelgangers, their own non-national territory. Navi Mumbai is a Shenzhen double of Mumbai. New Songdo City shadows Seoul. Surpassing irony, in Kazakhstan, Astana is a zone as national capital — a zone representing the state from which it is purportedly exempt, filled with paleo-Genghis imagery cooked up by famous architects: the pyramidal Palace of Peace and Reconciliation, or the gigantic microclimate tent that houses the Khan Shatyr entertainment centre. With the zone, the state can design a trap door out of its own laws and a proxy to engage in undisclosed and potentially lucrative dealings. ("Jawdropping", in TED-speak.)

A second vantage point with a good view of the giant is the global urbanism of broadband communications, and it is also often the easiest place to spot the remote or indirect forces that we have not trained ourselves to see. In 2000, there were less than 800 million cell phones in the world. By 2010, there were over 5 billion, and a majority of them were in the developing world.[1] Broadband is written into the platforms of national governments and into the development goals of international organisations like the World Bank and the UN. Access to mobile telephony — what the World Bank has called "the world's largest distribution platform" — is treated as a right, akin to the right to water or food.[2] New entrepreneurs identify multipliers and borrow crowd-sourcing techniques to penetrate the market in densely populated countries that are experiencing staggering increases in mobile telephony growth. In those countries where the phone is the internet link, Web 2.0 is changing farming, banking, medicine and education with corresponding

changes to urban arrangements. In a country like Kenya, one of the last to receive international fibre-optic submarine cable, the position of that fibre is hotly contested. It is not clear whether it will reinforce existing urban development on a corridor between Mombasa and Nairobi, generate zone-like enclaves, or penetrate rural areas to sponsor development and education. A state or any of the ballooning numbers of non-state players (such as NGOs, service providers, multinational enterprises, regulatory agencies), in any combination, may control this infrastructure space and create a monopoly or bottleneck within it.

With both zone and broadband urbanism, the overt policy declarations are all about free market, free trade or open access, but the arrangements on the ground are the work of a wilder Leviathan for which we have no studied political response. The zone remains under the radar, avoiding laws and side-stepping political declarations that might add friction to the otherwise lubricated condition. Broadband urbanism is a fluid game of interdependent players who can concentrate power or simply remain camouflaged within a complicated constellation of governance. There is no correlation between these consequences and published intent. Like a smuggling ring in which a number of people carry out chores but leave behind no sense of the entire organisation, there is a currency of actions decoupled from policy declarations — a currency of actions that can even outpace law. The result is a cell phone that still costs too much in Nairobi, or a world that seems composed primarily of segregated exurban formations, but it is not entirely clear why. This is the giant's magic, or sleight of hand. The question is not just, "who is making the rules?" It is something more like, "Who or what is making the milieu in which some things are possible and others are not?" Even if the giant can be detected despite its ephemeral activities and relationships, the question remains, how can it be manipulated, and what are the political implications? Important clues reside in the spatial arrangements themselves, but it is extremely difficult to locate the toggles and dials that created them.

After TED, there will be a trip to Davos in it for our Hugo. To the beard, body mic and publicist, add a pair of loafers and there could even be some expensive seminars where, for a price, our Hugo would explain what he really meant by the idea that "architecture is information" or "infrastructure is a medium of polity".

Activity in infrastructure space is information, or "the action is the form".

Evidence of the giant can be found in the field, but it can also be detected by adopting a new way of thinking. Marshall McLuhan's "the medium is the message" set out new territory by provoking just such an altered habit of mind. It was repeated so often that everyone at least pretended to know what it actually meant. Maybe Hugo's "architecture is information" is, similarly, a cultural meme sufficient to hold the idea of this new giant and allow it to circulate. McLuhan wanted to foreground, not the content, but the behaviour or repertoire of each medium from print to radio to TV. Content was the *"juicy piece of meat carried by the burglar to distract the watchdog of the mind"*.[3] We parse the world with minds trained to name and declare. Only the content, the spoken story on the radio, is palpable. The behaviour of the radio itself and the way it organises its listeners is harder to name: it is as if we can only know about the stone in the water but not the water.

Understanding the idea that "architecture is information", or making palpable the water, requires a mental journey similar to recognizing the behaviour of the radio, but it must also overcome associations with the word information. Information, especially in digital culture, is text or code — things that come up on screens to be interpreted through one sort of language or another. The more ubiquitous these devices become, the harder it is to see spatial technologies and networks that are independent of the digital. The world has become an "internet of things" — an interplay between smart buildings, smart cars and countless mobile phones and computing devices. Almost every discipline in the 20th century was in the thrall of information science, bent on determining and quantifying cybernetic *systems* of information with some degree of predictability. Architecture was among those disciplines, as the work of Cedric Price or Christopher Alexander and others might demonstrate. Late 20th century gurus like Kevin Kelly, celebrating the success of digital capital, ask culture to imagine cars as "chips with wheels", airplanes as "chips with wings, farms as chips with soil, houses as chips with inhabitants. Yes, they will have mass, but that mass will be subjugated by the overwhelming amount of knowledge and information flowing through it."[4]

Yet for some mid-20th century cyberneticians, those who foretold the digital revolution but who were not yet surrounded by all of its products, it was perhaps easier to understand that anything — dumb, inert, human, non-human, non-digital — could be a carrier of information and that the physical arrangement of infrastructure space is itself information. The social scientist and cybernetician Gregory Bateson saw information as the elementary particle of exchange in the practices of tribes in New Guinea, a meeting of Alcoholics Anonymous, or the communication of dolphins. (He theorised that the clicks of dolphin language were like ones and zeros.) Bateson would have no trouble seeing activity and information exchange in inanimate objects, and he would also speculate about the temperament or political bearing inherent in their arrangement. "Information is a difference that makes a difference", he famously wrote.[5] He used the example of a man, a tree and an axe as an information system. There is nothing "supernatural" about this power.[6] The objects do not need to be enhanced by digital technologies or coated with sensors. To the degree that they "make a difference" in the world, they create influence, intention and relationship that constitutes *information*. The information manifests, not in text or code, but in *activity*. For Bateson, any of the world's infrastructure spaces might be — like the man, the tree and the axe — producers or organisers of information. In some ways, his perspective is all that is

needed to understand Hugo's claim that "architecture is information".

Still, the idea that information is carried in activity, rather than systems of code, must struggle against cultural habits. How does one assess activity or information exchange in the static arrangement of familiar spatial infrastructures like a highway system, an electrical grid or a suburb? Spaces and urban organisations are usually treated, not as actors, but as collections of objects or volumes. Agency might be assigned only to the moving cars, the electrical current or the inhabitants. We are less accustomed to seeing that activity also resides in the relationship and relative position of the various parts of the organisation.

Yet, looking more closely at the familiar field of mass-produced suburban houses, the organisation embodies a distinct activity that is very apparent. The developer is not making 1,000 individual houses, but a kind of agriculture of houses — 1,000 slabs, then 1,000 frames, 1,000 roofs and so on. The house, as seen in pictures or stitched into needlework, is akin to McLuhan's content; it distracts us from what is really being made. The field of houses is enacting a propensity to organise all activities across a population of houses. It privileges these repetitive activities and renders the act of making an individual house into a marginal gesture. What is really being made is something like a protocol or a non-digital spatial software that is both shaping and generating the activity of making houses. The relative changes in this organisation are, as Bateson would say, "differences that make a difference". The organisation is *doing* something, and changes within it constitute information. If we focus only on the house, this larger process remains as a kind of ghost or ghostly giant in the background. The architect who has only been trained to make enclosures will always rush to design the single house, so as to have something to show, only to be outwitted by the equivalent of McLuhan's media or Hugo's reanimated giant.

While the activities of an organisation may remain undeclared, we can still look squarely at what *is* declared: at the scripts, stories and promotion it produces. A script may simply direct the use or application of a technology, as in the choice to use electricity for lighting. A script may also set the ideological course for a technology or become the promotional story that serves as content — like the Cape Cod cottage or the Arnold Palmer Village. Highways have historically been associated with redemptive qualities or political destinies such as freedom, democracy or patriotism. The free zone is associated with openness and streamlined bureaucracy — a one-stop shop for global business. In broadband infrastructure, mobile telephony is associated with prosperity and open access to global networks of information. In any infrastructure network, the script may be the thickest strand — virtually bending and shaping the technological instrument. Examining these scripts helps to make clear the activity that either exceeds them or is discrepant from them, like the developer agriculture that overwhelms the individual house.

The giant of infrastructure space might be analyzed in this way, foregrounding the medium of activities that remain unnamed but are nevertheless consequential. This space composed of action does not render form-making impossible but rather points to an additional mode of form-making with special powers. Forms composed as activity (the agriculture houses) shape a population of forms composed as objects (houses), in accordance with certain scripts. If architects are often making a stone in the water while the world makes the water, the stone is an object form while the water is what might be called the *active form*. A better McLuhanesque meme might be: the action is the form.

Infrastructure spaces are performers or active forms.

When Hugo's celebrity gambit works, the notion that "the action is the form" could be an attractive subject for the popular essay as detective story of ideas (or what we might call *The New Yorker* treatment). These treatments usually follow a similar formula. A narrator, writing in the first person, and with varying degrees of egotism and modesty, encounters a number of thinkers. Each encounter provides clues to the comprehension of an idea. Immediacy is essential. It is a crisp autumn day when the narrator meets the thinker who is really hard to understand. Or the narrator meets the thinker who is really hard to understand in an airy New York apartment, and the airiness of the apartment helps the reader to finally comprehend global finance. Or the narrator sits across from the particle physicist, who has tousled sandy hair that is parted on the side. With descriptions such as these, the reader has a better chance of understanding things like string theory. It really works.

The sociologist Bruno Latour has dark hair that is parted on the side. He speaks with a French accent. His studies amplify the notion that "the action is the form". In what he calls Actor Network Theory (ANT), he proposes that socio-technical networks, like infrastructure space, are created by both humans and non-human technologies and that the technologies themselves are actors or "actants" in this process. Actants are "doing something".[7] Technologies influence the desires of social networks that reciprocally shape them: humans develop the computer but the computer, in turn, changes the way that humans think. Humans design the technology for the suburban field of houses and that environment, in turn, shapes human interaction. By studying what a technology is doing, Latour studies not only the declared story or script but also the activity in an organisation — what the infrastructure is saying as well as what it is *doing*. Latour argues that this interaction between script and technology is indeterminate, like a flow of information or a flow of water. He variously describes action as a "surprise" or "mediation".[8] It is "borrowed, distributed, suggested, influenced, dominated, betrayed, translated".[9] Social forms are not something to be taxonomised and fixed. They are "under-determination" in an ongoing process.[10] To determine this information, to give it a name or boundary, is to negate it or stop the flow.

Turning to theatre, where constructing indeterminate activity is a completely ordinary or practical matter, Latour writes about the use of the word "actor" in social studies:

> "It is not by accident that this expression, like that of 'person', comes from the stage … Play-acting puts us immediately into a thick imbroglio where the question of who is carrying out the action has become unfathomable."[11]

An actor adheres to an explicit script, but the scripted words are considered only to be traces or artefacts that provide hints of an underlying action. While performing, actors rarely deal with states of being that can be named. Actions are the currency or the carrier of information. An actor would not play "being a mother" but rather "smothering a child". Actors are also at ease with discrepancy, the way that action escapes text to remain indeterminate. The character saying, "I am pleased to meet you" may actually be expelling someone from society. The character saying, "I don't love you" may actually be straining to connect. Action cannot be declared and it can be decoupled from declaration. Finally information about the character emerges from a string of actions. Similarly, infrastructure space is performing, and the changing shape of that stream of activities constitutes information.

In the interplay between scripts and technologies, this activity seems almost to emerge as a third thing. Looking past the declared script in infrastructure space, the undeclared or decoupled activities emerge more distinctly. In the suburban field of houses, the buyer of the Colonial Cape

Cod dwelling or the Arnold Palmer Villa is buying a script, when what is really being delivered is a development process that remains undeclared. A story about decentralization may accompany an electrical utility or mobile telephony network while its arrangement actually allows it to harbour monopolies. DARPAnet, scripted as a stealth network, becomes the internet commons. The free zone, heralded around the world as the instrument of openness and anti-bureaucratic trade networks, creates new forms of bureaucracy. Facebook, scripted for innocuous social networking on a college campus, unfolds in the Arab Spring as an instrument of dissent.

In these cases, information resides not only in the script or the technology alone but also in some immanent activity or capacity that escapes explanation — what we might call the *disposition* of the matrix. It is information that is not delivered by means of the familiar mechanisms of text and code. This undeclared *disposition* that remains as a ghost to declared script has perhaps lent some unwarranted mystery or magic to the giant.

Dispositions and active forms of infrastructure space can be designed.

While he was alive, philosopher Gilbert Ryle spoke with a British accent, smoked a pipe and wrote philosophy in a droll, charming and conversational tone. He is perfectly cast as the next encounter in *The New Yorker* treatment of "the action is the form". Ryle was especially keen to point out the "ghosts in the machine" or the logical fallacies harbouring in everyday language. Disposition was, for Ryle, one of those ghosts, and his work further demystifies the magic of ghosts and giants with a practical art that might also guide the design of infrastructure space.

Ryle wrote about disposition as something we already understand and use in common parlance: an unfolding relationship between potentials — a tendency, temperament or property in either beings or objects — a propensity within a context. The disposition cannot be proven as a definite "occurrence". Ryle used the example of glass that does not need to shatter to possess a brittle disposition. (Like Latour and Bateson, Ryle considered dispositional qualities in both human and non-human subjects.) The shattered glass is not a "ghostly happening, but because it is not a happening at all".[12] A ball on an inclined plane possesses disposition that is stored in relationship and geometry.[13] The ball does not need to roll down the inclined plane to possess the disposition to do so. A function in calculus is an expression for the behaviour of a number of values; knowing all of those values is less important than understanding the disposition of the function to form a curve with a particular amplitude. Reinforcing Latour, Ryle argues that, given this latency, disposition is indeterminate. Yet this indeterminacy is not necessarily mysterious. Ryle wrote:

> "My being a habitual smoker does not entail that I am at this or that moment smoking; it is my permanent proneness to smoke when I am not eating, sleeping, lecturing or attending funerals, and have not quite recently been smoking."[14]

For Ryle, the difference between "knowing that" and "knowing how" — between training the mind to know the answer over training the mind to rehearse actions — was essential to an understanding of disposition. One cannot know disposition as one knows the correct answer. It is gleaned from multiple observations of activity. The disposition of the organisation is an indication of how that organisation deals with the interplay of factors over time. Like Latour, Ryle refers to theatrical arts that are accustomed to handling action. Using the performance of a clown, Ryle noted that being funny is not something for which one knows the right answer. It is something the comedian knows how to do in an unfolding response with an audience. "Knowing how" is dispositional.[15] Ryle noted that dispositional attributes sometimes remain as a fuzzy imponderable within customary logics and language structures — "occult" agencies or processes taking place, in "a sort of limbo world".[16] Yet to disregard "knowing how" in favour of "knowing that" is to discount our most practical expressions for capacity, potentiality, property or tendency. Best understood by using it, the word "disposition" is itself dispositional.

Ryle never addressed disposition in urban space, but if he had he might have talked about the disposition of the suburban field to multiply or the disposition of Facebook to become a political instrument. He might have referenced simple topologies or familiar network repertoires that have a disposition to circulate information in particular ways: the phrase "a smuggling ring" models the secrecy of a closed loop of players. The disposition of a linear rail system or a linear fibre-optic cable is different from that of an atomised sea of mobile telephones. A radial or hub and spoke network, like mass media television or radio, is one in which every activity or relationship must contact a single central point with the privilege of disseminating information. A tree or arborescent structure is hierarchical and concentrates authority, like streets that move from trunk

line arterials to the smallest lanes. An all-channel or distributed network, like an open mesh, is one in which every point in the network can reach every other point. A mainframe computing network is a linear network that passes information sequentially, while a parallel network is arranged with simultaneous rather than sequential exchange. A skyscraper might be understood as a serial organisation because one accesses each floor sequentially by an elevator. A market, a train station or any other organisation with multiple points of access and exchange might be organised with simultaneity and parallelism. Disposition can then be designed, just as one can decide the geometry and relative position of the ball on the inclined plane.

If Bateson joined this conversation, he would even see, in these simple topologies and relationships, disposition as a marker of political temperament. He assessed network architectures for the ease with which information flowed within them. He theorised that some binary relationships tended toward competition, violence and instability (eg twin daughters vying for a parent's love, two nations warring over territory, alpha dogs and beta dogs). On the other hand, reciprocal relationships (eg organisations where actors take turns being dominant or submissive) were more stable. Extending Bateson's theory, a serial organisation does not circulate information as robustly as a parallel organisation. The parallel organisation is, at once, more open and more stable. Similarly, a hub and spoke or an arborescent network can concentrate information and authority. The means by which information flows can be indicators of aggression, submission, duplicity, violence or resilience immanent in an arrangement — indicators of political bearing that are partly responsible for the power that infrastructure space is able to exercise. The zone, which, like a smuggling ring, is structured as a closed loop, possesses a *disposition* to ignore dissent or any information incompatible with its business platform. In studying the formation of broadband urbanism, we could compare the linear topologies of fibre-optic cable with the distributed population of cell phones, to see how each possesses a different *disposition* to territorialise and concentrate power.

Bateson, Latour and Ryle offer techniques for designing the water. They deepen an understanding of the disposition of active forms, which are not single events or objects but something more like variables in a flow of information. They analyze the performance of infrastructure space — now no longer the ineffable background or elusive political medium. However complex the giant of urban infrastructure, these thinkers offer some simple toggles and faders to begin to manipulate it.

Making active form and disposition is, for architecture, an extra art.

Our new Hugo brings his message to a non-architectural audience, in part, because if space is a secret weapon of the most powerful people on earth, it is a secret that has perhaps been best kept from architecture. For designers, authorship of form is often limited to object form — the aesthetics of which coalesce around connoisseurship of profile and geometry. While the discipline considers activities like programme, the idea of active form is seemingly oxymoronic or mystical. If summoned to create it, architects would naturally rely on what they are best trained to create — a formal object *representing* action. A single enclosure would be crafted to represent, for instance, relativity or dynamism. A construction might be coated in digital sensors that respond to movement, touch or programming. A more simple-minded confusion (made more powerful by being simple-minded) arises when action or activity is confused with movement or kineticism. In a spatial field with evolving components, the architect might design the entire *field* with a fixed architectural pattern. The more dynamic or agitated these tracings, the more "active" the form is meant to be.

The distinction between understanding form as object and form as action is something like Ryle's distinction between "knowing that" and "knowing how". The importance of active forms lies in what they, for instance, inflect, ignite or suppress. They establish a set of parameters or capacities for what an organisation will be *doing* over time. Active forms might describe the way that some alteration performs within a group, multiplies across a field, reconditions a population or generates a network. The designer of active forms is designing not the field in its entirety but rather the delta or the means by which the field changes — not only the shape or contour of the game piece but also a repertoire for how it plays.

The art of making active form presents a non-modern opportunity. Hugo might now observe that his original "this kills that" construction — the notion of progressive succession that propels the utopian or the *avant garde* — turns out to be a mistake in thinking. ("Doom loop", he explains, wearing loafers.) Active form does not need to kill object form to exist. The disposition of that sort of argument is one of Bateson's tense binaries. There is no need for competition between these skills, but rather the recognition of a broader category that includes both object form and active form. The two coexist in excess of each other. Active form only redoubles design capacities offering additional techniques and artistic pleasures — extra arts with abundant cultural power. The toggles of active form are always, and have always been, present in any design. It is an artistic choice whether or not to use them — whether to place object form in relation to another set of potentials — like the ball on the inclined plane. Active form may partner with and propel object form, determining how it travels through culture with time-released powers.

Most artists and entrepreneurs in the world insist on a repertoire that includes both object form and active form. A geneticist cannot represent all the gene sequences of DNA with an image of a double helix but can engage the ongoing development of an organism with an active form that alters one of those gene sequences. A computer scientist would never attempt to fully represent the internet but would rather author active forms that ride the network with very explicit instructions. An entrepreneur designs not only the product but also its passage through a market. An environmentalist would not attempt to work on a forest by placing every bird in every tree or planting every sprig of undergrowth, but would rather send in a bird or an insect or a seed to recondition the landscape over time.

Along with plan, section and elevation, the designer can assess a dispositional dimension. In the field of suburban houses, one could design a single house as well as use the repetitive organisation

as a carrier or multiplier. Just as the elevator was a germ of the skyscraper morphology of cities, a contagious detail or technology (eg structural innovation, skin detail, vehicle) can move through a population of buildings. Object form can also become active form when it acts as a switch or a stone in the water. By crafting relationships, the designer can design interdependency and even design indirectly or remotely, changing A to change B to change C. The designer of active forms is perhaps designing not just an enclosure or an urban boundary but a disposition for the growth or even contraction of space. The author of software is not the author of all of the forms that flow through that software, but rather the author of a platform that shapes and manages information. Like a calculus function — an explicit expression like cosx — software provides variables that facilitate a relationship while not controlling every outcome. A software is, like the formulas for making spatial products or free zones, a form for handling forms. A designer can write the update or the killer app (that never actually kills).

In this expanded repertoire of form-making, there is an art in knowing when it is most powerful to be the stone in the water and when it is most powerful to be the water. Both have material consequence. The Invisible Man was only powerful and sneaky because he both appeared and disappeared. When the man himself was not visible, the space that he disturbed was visible. A drink was drained from a glass, doors opened and closed. Architecture has the same opportunity to create enormous spatial consequence by occasionally becoming its own medium — by becoming infrastructural. Infrastructural space can direct changes to object forms by designing the flow of actions and variables. The action is the form. These are the extra arts of making active form and disposition — magic one can only know *how* to do.

Hugo's new audience needs spatial studies.

The new audience for Hugo's TED talk — the audience already persuaded about the political magic of infrastructural space — is composed of social, political and business entrepreneurs. Yet while perhaps better networked and more powerful than architects, an understanding of space is a missing ingredient in their conversations.

Spatial arrangements are often the inadvertent outcome of rules written in the jargon of business, real estate, logistics, trade, banking, informatics or governance. For instance, the free zone — an outmoded formula for establishing sub-economic conditions in an exurban enclave — is the result of trade laws, development strategies and economic policy. In the new urbanism of broadband, there is a growing expertise concerning ICT (information and communication technology) and its direct effects on development, but for global consultancies like McKinsey and others, this intelligence is managed with "econometrics". Strangely, there are very few development expressions that originate from a *spatial* analysis.

Many sciences look at infrastructure: social sciences, economics, informatics, science and technology studies, the history of science, organisation studies, media and communication studies, architecture and urbanism. Yet the most innovative thinkers in all of these disciplines — those making up our Hugo's new audience — often now insist on stretching their respective disciplinary habits beyond theoretical purity or beyond the assumptions and authorities of science to consider their subjects in a more complex context with multiple actors, processes and events. They are, arguably, attempting to expand the factors considered in their study to include more circumstances delivered by uncharted but consequential spatial practices.

For instance, Esther Duflo, an innovative economist studying poverty, has developed techniques for testing economic theory against a variety of small contextual factors on the ground. The trials demonstrate that small incentives or cultural signals attached to such things as immunizations or the distribution of mosquito nets finally make the difference between the success or failure of a campaign to relieve hunger or disease. Richard Heeks, director of the Centre of Development Informatics at the University of Manchester, has suggested that rather than consider econometric indicators alone in policy-making decisions, it is important to look at economic, socio-cultural and legal ramifications.[17] After seeing the failure of some economic science on the ground, Heeks has found it fruitful to consider Latour's Actor-Network models — setting aside some scientific assumptions to follow a fluid process with multiple players.[18] Over mathematical assessment and measurement, Heeks proposes more qualitative and analytic markers. He tracks the entrepreneurs of new crowd-sourcing cell phone platforms for banking, agriculture, medicine or marketing. He asks, "Where are the Amazon and eBay of international development? What would be a 'Development 2.0?'"

Since architecture and urbanism will be a major determining factor of population concentrations and the accessibility, resilience and disposition of any global network, the question might be, "What is space 2.0?" While software for digital environments, and development formulas for spatial environments like the zone draw more and more of the world into common platforms, an entrepreneur would see opportunity in the startling lack of resilient spatial formulas.

The zone format, for instance, is the functional equivalent of an early, stand-alone word processing programme waiting to be joined by other softwares. The spread and mutation of this zone, despite its failures, demonstrates the power of spatial platforms even in the absence of their fitness. One can design an object form for the zone by designing another glittering skyscraper, which would have its own merits, experiments and beauties, but one can also manipulate the

active forms in this complex. The zone is itself a germ, like one of the houses in the suburban field — the potential multiplier of a new technology, programme or political intention for building that exceeds the single enclosure. The zone's position as an exurban enclave enhances its exemption from law or other political inconveniences. Nevertheless, as the zone continues to swallow the city, a crafty designer might find opportunity in manipulating its disposition instead of its morphological outlines. One could deliver the same technologies and scripts together with a different disposition by, for instance, mapping the zone onto the city itself — the city that the zone previously departed for an exurban enclave. Incentives and security remain the same, but the disposition of the organisation would go from something like a closed loop to an all-channel network. HITEC City could be mapped on to Hyderabad, or the Athi River zone could be mapped onto Nairobi. Urbanity would lend patency and extra players to make information exchange more robust. If this alteration of the formula were as contagious as any other permutation of the zone has been, the shift in disposition could have enormous, ramifying consequences.

While urbanists frequently analyze urban values and morphologies associated with physical infrastructures such as rail, highway, power and water utilities, the discipline is under-rehearsed in an analysis of the compound spatial dispositions attending broadband infrastructure. Linear fibre-optic cable privileges the territory through which it runs. The same fibre also provides the necessary bandwidth for contemporary mobile telephony, a technology with an atomised topology that promises to create ubiquitous coverage. The service providers who sell and transfer broadband between these overlapping organisations add a third dispositional factor by creating point sources of access — a switch — in the constellation. Any design might, to some advantage, continually reassess the *disposition* of the network as it grows, paying attention to what it is doing as well as what it is saying.

Spatial dispositions, while rarely discussed in development strategies, might reasonably be the crucial medium of any economic or informatic endeavour. If space is often the *defacto* medium of polity, to handle this space is to handle the raw, consequential material of politics. Architects, with their understanding of organisational structure and the disposition immanent within it, have skills to match the giant's magic. There are no prescriptions that can be named, no correct answers to cure problems — sometimes not even politics that can be declared. Again, manipulating disposition is about *knowing how* not *knowing that*.

The giant is a political instrument.

Hugo's new giant might seem, to the classic activist, to be the old Goliath. To counter authoritarian forces, dissent, as resistance and refusal, must often assume a binary, oppositional disposition. Yet just as the architect may discover that they are making object form in a context that only responds to active form, so the activist frequently discovers that their targeted power has escaped. The powerful players project themselves with proxies and obfuscation. They keep a currency in untraceable actions and relationships. Activism shows up at the barricade, while Goliath moves his headquarters to a free zone. Goliath even finds a way to pose as David and, in costume, wanders away from the bull's-eye. A little epidemic of rumour and duplicity finally captures the world's attention better than forthright political expression. Indeed, the notion that there is a proper realm of political negotiation usually acts as the perfect camouflage for the giant's more elusive behaviour.

Yet two can play at this game. To learn the extra arts of active form and disposition is to learn to exploit the giant — not in a binary fight but an ongoing political dissensus. Unfolding, indeterminate and never cured with a single revolution, dissensus aligns with and deploys a dispositional habit of mind. Changes to disposition, like changes to a medium, can create broad foundational sea changes or domino effects. While those political traditions that call for inversions and revolutions often call for the absolute annihilation of the preceding system, dispositional shifts are part of a sustained reconditioning or revolutionizing of a spatio-political climate. Dispositional shifts may also adjust political temperament to preclude binary symmetrical fights and reduce the tensions and violence that attend them. Discrepancy between what an organisation is saying and what it is doing is a signal marker for disposition that escapes declaration. It also lends to space a capacity for duplicity. An organisation can be a sly agent and a shrewd performer. The rational, righteous cornerstones of dissent are often finally less instructive or consequential than duplicity or fiction. Discrepancy is an important engine and a different, perhaps better, tutor of alternative political arts.

A special notion of aesthetics attends these alternative political arts. In *The Politics of Aesthetics*, philosopher Jacques Rancière outlines an aesthetics that "does not refer to a theory of sensibility, taste, and pleasure for art amateurs". Rather, he focuses on "aesthetic practices" that both "depict" and enact, that articulate "ways of doing and making". Aesthetics exists as changing regimes of forms that are full of meaning but not determinate meaning.[19] Significantly, Rancière writes not about the aesthetics of politics, but the politics of aesthetics — the politics surrounding the reception of a work of art that may be discrepant from declared intentions. For instance, he writes about Flaubert's *Madame Bovary*, a novel that, in reception, relayed to its audience a liberating disposition despite Flaubert's conservative politics. One can think of other stories when script and political disposition were decoupled or when the action was the form. Bob Dylan's *Like a Rolling Stone* was a bad-boy song that became an anthem of the counter culture movement. Since it is very easy to demonstrate that Barack Obama is Christian, saying that he is Muslim is a very effective rumour: the rumour can be kept alive even longer and repeated twice as much, first to spread the falsehood and then to refute it. As important as the content of the rumour is its dispositional active form — the way it behaves and bounces.

For architecture, an aesthetic training in these political arts might resemble that of theatre. Those in the theatre come very close to handling action as their primary raw material. Working up to their elbows in it, they also understand its discrepancy from text. Just as actors have recipes and tricks for fooling their body and voice into being another body and voice, their job is to create mixtures of opposing intentions — playing actions that are entirely different from outward movements, text and gestures. Sociologist Erving Goffman, who was interested in discrepant

characters as well the discrepancies in every person's everyday performance, speculated that a training in duplicity is hard to formalise. Most of the techniques can only be "picked up", as in the training of a shill, a hustler or a confidence man.

The performance of active forms and object forms together and an understanding of their inherent discrepancy enhances the political power of architecture. Going still further, the art of manipulating the giant not only demystifies it but also uses its supposedly formidable size to advantage. If it were possible to tinker with fables and stories, might Tom Sawyer slip into David's role? Rather than going to the effort of killing Goliath, an activist can leverage the giant's power and size, or use it as a multiplier in a way that is more manipulative than collusive. A refreshed activist repertoire learns from a number of similar slippery characters — pirates, princes, prisoners, comedians and celebrities. Overt righteous activism is sometime warranted. Yet sometimes a better strategy might be to turn 90 degrees just before the finish line or, never turning around for the duel, to continue pacing away from one's opponent into a fresh territory of opportunity.

This extended dispositional repertoire is less self-congratulatory, less pure, less automatically oppositional, more effective and sneakier. Deploying active forms in the calculus of disposition, an activist might learn to use techniques like gossip, rumours, multipliers, gifts, compliance, misdirection, distraction, meaninglessness or entrepreneurialism. These techniques do not belong to the proper politics of either the left or the right. They are the unnamed but consequential forces of infrastructural space, and they can excite feelings of resourcefulness and ingenuity. Infrastructure, with all of its multipliers, can spread change like gossip, and fictional scripts, like those that distract from the real activities of the free zone, can also be used towards alternative political ends. Dispositional techniques (especially the sneaky ones, like exaggerated compliance or gifts that cannot be refused) may refuel the architect's traditional role in making object form, but they also open onto a vast pasture of alternative practices in which the architectural entrepreneur is writing the non-digital software for how an urban space *performs* over time. These explicit but indeterminate expressions — a cosx for spatial relationships and interdependencies — can, for instance, overwrite the free zone or recondition the space of broadband urbanism. Designing these expressions, not as a developer or economist would, but as only an artist and architect can design them, enhances the discipline relative to these other players. Compounding that power is the fact that what architects have running through their fingers is already a medium of polity, the design of which is capable of relieving abuses or outwitting authoritarian forces.

Hugo's performance, like the organisations he describes, is enacting his position. Like all of the other shills, comedians and confidence men, he is *demonstrating* to his audience that the technologies of infrastructure space are in a consequential position. Similarly, with a little handling and with techniques that are "picked up", spatial entrepreneurs can game space with — not a science — but an art of infrastructure change. Like the Invisible Man, this entrepreneur knows something about the spatial consequences of both appearing and disappearing. Architecture exercises its powers by designing not only the stone in the water but the water itself — by occasionally dissolving into its active forms and becoming information. It's the disposition of the performance that matters — who says it, how they say it, who repeats it and to whom. The action is the form.

Notes

1 “A 2010 Leadership Imperative: The Future Built on Broadband,” (ITU, The Broadband Commission for Digital Development, 2010).

2 Mohsen Khalil, Philippe Dongier, and Christine Zhen Wei Qiang, “Overview,” in *Information and Communications for Development: Extending Reach and Increasing Impact*, ed. World Bank. Development Data Group. and World Bank. Global Information & Communication Technologies Dept. (Washington, D.C.: World Bank, 2009).

3 Marshall McLuhan, *Understanding Media: The Extensions of Man* (New York: McGraw-Hill; London: Routledge & Kegan Paul, 1964, 2001), 19. The quotation: “For the ‘content’ of the medium is like the juicy piece of meat carried by the burglar to distract the watchdog of the mind.”

4 Kevin Kelly, *New Rules for the New Economy* (New York: Penguin Books, 1998), 76.

5 Gregory Bateson, *Steps to an Ecology of Mind* (Chicago: University of Chicago Press, 2000), 381, 462, 315, 272, 21.

6 Ibid., 472, 464.

7 Ibid., 46.

8 Ibid., 45.

9 Ibid., 46

10 Ibid., 44.

11 Bruno Latour, *Reassembling the Social: An Introduction to Actor-Network Theory* (Oxford: Oxford University Press, 2005), 46.

12 Ibid., 43.

13 Francois Jullien, *The Propensity of Things: Toward a History of Efficacy in China* (New York: Zone Books, 1995), 29.

14 Gilbert Ryle, *The Concept of Mind* (Chicago: University of Chicago Press, 1949), 43.

15 Ibid., 27–32.

16 Ibid., 119–120.

17 Richard Boateng et al., “E-Commerce and Socio-Economic Development: Conceptualizing the Link,” *Internet Research* 18, no. 5

(2008), www.emeraldinsight.com/1066-2243.htm.

18 Richard Heeks and Carolyne Stanforth, "Understanding E-Government Project Trajectories from an Actor-Network Perspective," *European Journal of Information Systems*, no. 16 (2007).

19 Jacques Rancière, *The Politics of Aesthetics* (London: Continuum, 2004), 22–23, 13–14.

About the author

Keller Easterling is an architect, writer and professor at Yale University. Her book, *Enduring Innocence: Global Architecture and its Political Masquerades* (MIT, 2005) researches familiar spatial products that have landed in difficult or hyperbolic political situations around the world. A previous book, *Organization Space: Landscapes, Highways and Houses in America*, applies network theory to a discussion of American infrastructure and development formats. A forthcoming book, *Extrastatecraft: Global Infrastructure and Political Arts*, examines global infrastructure networks as a medium of polity.

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